UNC Charlotte

ECGR 3120 Electromagnetics

Magnetically Steered Cardiac Ablation

Milestone 2: Force and Torque / Catheter Steering

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Input status: The approximate $0.013\,\mathrm{A\,m^2}$ tip moment is from the project handout. The shaft stiffness used for the default plots is inferred from the approximate 60°-command/30°-tip baseline at an illustrative 80 mT; it is not an official or measured stiffness.

Abstract

This report models magnetic steering torque on a catheter tip and the counteracting torque of a flexible shaft. The dipole result is checked against a discretized equivalent-current-loop integral, and the equilibrium angle is solved from torque balance. The handout's approximate tip moment is used. No official shaft stiffness or complete baseline conditions were found in the starter workspace, so the runnable example infers an explicitly illustrative stiffness from the approximate 30° response to a 60° command at an assumed 80 mT. The resulting plots are demonstrations of the model, not verification of official baseline data.

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1 Objective

The objective of Milestone 2 is to estimate how much torque the magnetic field applies to the catheter's permanent magnet and what catheter tip angle results when the shaft's restoring torque is included. This calculation connects the magnetic field to a mechanical steering response. Angles are calculated in radians and displayed in degrees.

2 Input status and assumptions

The handout provides an approximate tip magnet moment of

$$m = 0.013\,\mathrm{A\,m^2}.$$

It also gives an approximate 30° tip-angle baseline for a 60° command. The workspace did not contain a shaft stiffness, the complete conditions associated with that baseline, professor-provided Milestone 2 signatures, a stated torque tolerance, or `tests/test_m2_steering.py`.

To make the graphs runnable without presenting an invented shaft value as official, the default script infers a stiffness from the approximate baseline using an 80 mT field. The latter is an illustrative comparison field based on the Milestone 1 example, not a verified Milestone 2 input. The inferred stiffness is

$$k = \frac{mB\sin(\theta_{\mathrm{command}} - \theta_{\mathrm{tip}})}{\theta_{\mathrm{tip}}} = 9.931\,268\,45 \times 10^{-4}\,\mathrm{N\,m\,rad^{-1}}.$$

This value is a derived calibration, not a measured shaft property. The script's stiffness command-line option allows it to be replaced when the official value is known.

3 Governing equations

In the model's x-z steering plane, the dipole moment has magnitude m and angle θ relative to $+z$:

$$\mathbf{m}(\theta) = m\begin{bmatrix}\sin\theta \\ 0 \\ \cos\theta\end{bmatrix}.$$

The magnetic torque on a dipole is

$$\mathbf{T}_{\mathrm{mag}} = \mathbf{m} \times \mathbf{B}, \qquad |\mathbf{T}_{\mathrm{mag}}| = mB\sin\theta,$$

where the latter θ denotes the angle between the moment and field. For a current loop, the corresponding integral forms are

$$\mathbf{F} = I\oint d\mathbf{l} \times \mathbf{B}, \qquad \mathbf{T} = I\oint \mathbf{r} \times (d\mathbf{l} \times \mathbf{B}).$$

The shaft is modeled as a linear torsional spring:

$$T_{\mathrm{shaft}} = k\theta_{\mathrm{tip}}.$$

At equilibrium, the magnetic and shaft torques have equal magnitude and opposite direction:

$$mB\sin(\theta_{\mathrm{command}} - \theta_{\mathrm{tip}}) = k\theta_{\mathrm{tip}}.$$

The equilibrium solver returns the root in radians. A positive spring stiffness keeps the modeled catheter unbent when the magnetic field is zero.

4 Equivalent current-loop method

The verification loop is a circle of radius 25 mm carrying 1 A, represented by 512 directed straight segments. For each segment, the force contribution uses the segment vector $\Delta\mathbf{l}$ and field evaluated at its midpoint. The torque integral also uses the midpoint position from the loop center. Summing these terms approximates the continuous line integrals.

For a circular loop, its dipole moment is $\mathbf{m} = IA\hat{\mathbf{n}}$. The torque from the current-loop sum can therefore be compared independently with $\mathbf{m} \times \mathbf{B}$. The loop used for this numerical check is a verification model; it does not assert that the catheter's permanent magnet has this exact geometry.

5 Verification results

5.1 Net force in a uniform field

A closed loop in a uniform field has zero net force because the vector force contributions around the loop cancel. The 512-segment result was

$$|\mathbf{F}| = 8.281\,050 \times 10^{-19}\,\mathrm{N}.$$

This effectively zero value is consistent with floating-point roundoff.

5.2 Current-loop torque compared with dipole torque

For $\mathbf{B} = [0.060, 0.080, 0.020]$ T, the numerical current-loop integral gave

$$\mathbf{T}_{\mathrm{loop}} = [-1.57075690 \times 10^{-4},\ 1.17806768 \times 10^{-4},\ -1.44343017 \times 10^{-22}]\ \mathrm{N\,m}.$$

The dipole calculation using the moment of the same discretized loop gave

$$\mathbf{T}_{\mathrm{dipole}} = \mathbf{m} \times \mathbf{B} = [-1.5708 \times 10^{-4},\ 1.1781 \times 10^{-4},\ 0]\ \mathrm{N\,m}.$$

Their absolute difference was $1.092\,641 \times 10^{-19}$ N m, and their relative error was 5.564913×10^{-16}. No professor-specified tolerance was present. For this report, agreement is checked against a local relative-error limit of 10^{-10}; it is not represented as the professor's tolerance.

5.3 Equilibrium and field-strength checks

At 80 mT and a 60° command, the computed equilibrium was 30.000 000°. The torque-balance residual was 0.000 000 N m at printed precision. With zero field, the computed tip angle was 0°. Holding the command at 60° and the inferred stiffness fixed produced:

Table 1: Computed equilibrium for a 60° command

Field	Achieved tip angle	Command error	Interpretation
40 mT	19.488156°	40.511844°	Illustrative
80 mT	30.000000°	30.000000°	Baseline-calibrated
160 mT	40.350950°	19.649050°	Illustrative

The achieved angle increases as field strength increases, as expected for fixed magnet moment, command, and spring stiffness. The 80 mT case matches the approximate handout prediction because the default stiffness was inferred from that prediction. It is therefore an illustrative comparison and not an independent validation of the baseline.

6 Figures

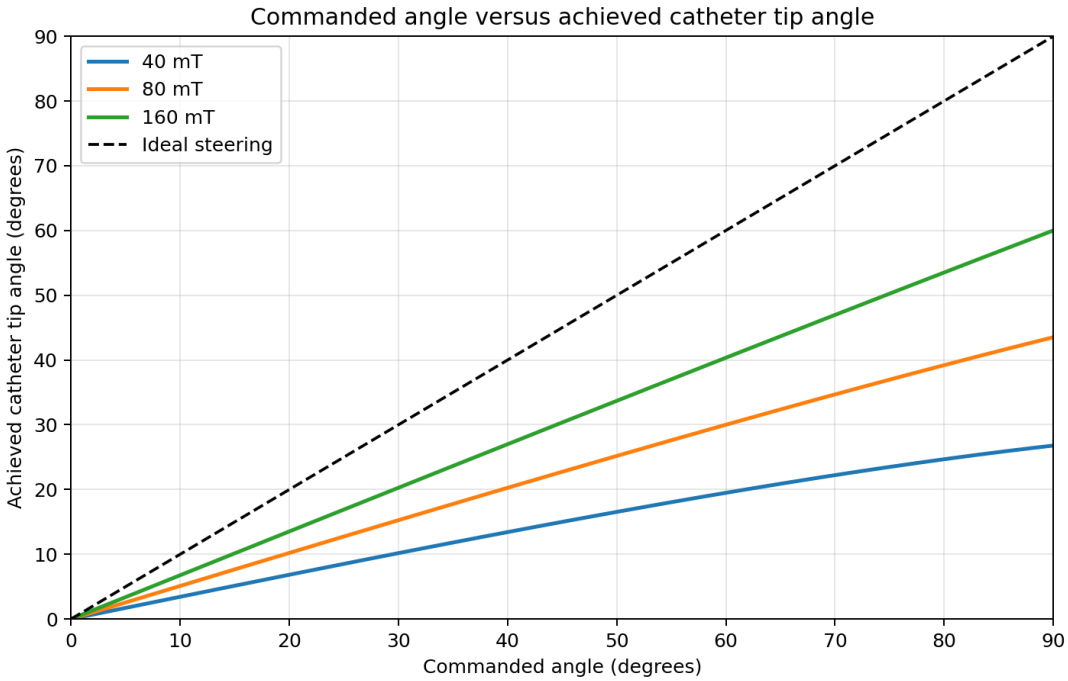


Figure 1: Computed catheter tip angle versus command for three field strengths. The dashed ideal-steering line is where achieved angle equals command. Stronger fields move the response toward ideal steering under the same spring assumption.

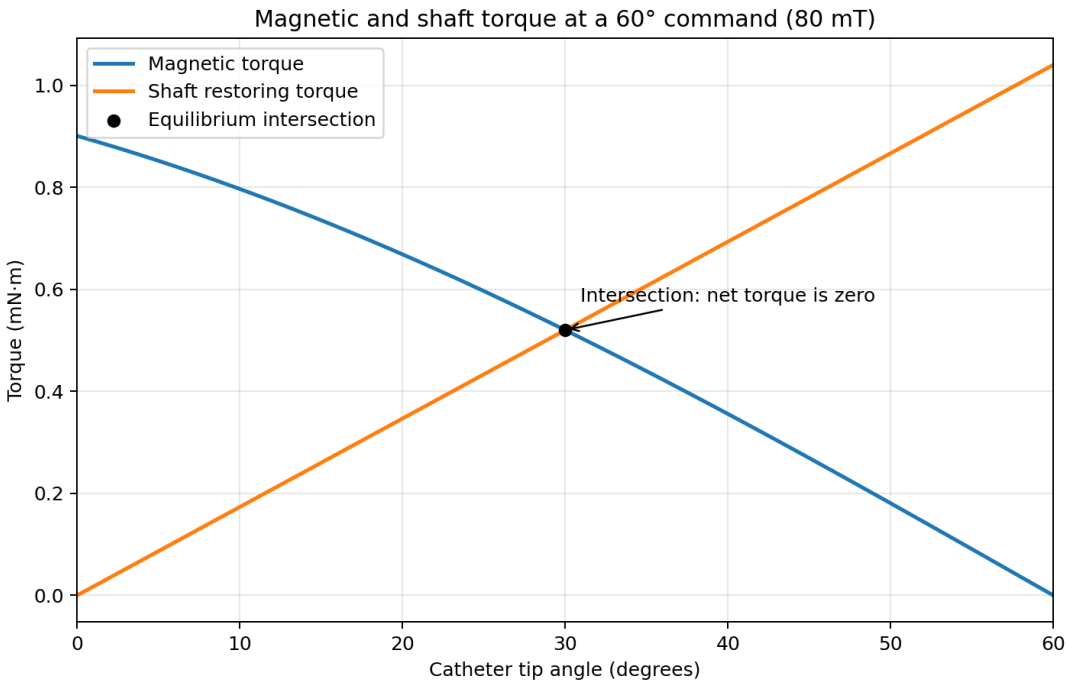


Figure 2: Magnetic and restoring torque versus tip angle for a 60° command and 80 mT field. The intersection is the equilibrium: magnetic and shaft torques balance, making net torque zero.

At the plotted equilibrium, the magnetic and shaft torques are equal and opposite:

$$T_{\text{mag}} = T_{\text{shaft}} = 0.520\,000\,\text{mN m}.$$

The calculated moment vector at a 60° orientation in the modeled plane is

$$\mathbf{m}(60^\circ) = [0.01125833,\ 0,\ 0.0065]\ \text{A m}^2.$$

7 Engineering interpretation

The magnetic field supplies steering torque, while the catheter shaft resists bending. Thus, a stronger field or a larger magnetic moment tends to increase the achieved deflection, and a stiffer shaft tends to decrease it. In this model, moving from 40 to 160 mT increases the predicted angle at a 60° command from about 19.5° to 40.4°. The calculation gives an understandable first-order relationship between electromagnetic design and catheter steering, but it does not establish clinical performance.

8 Limitations

The field is treated as uniform at the tip, the magnet as a fixed dipole, and the shaft as a planar linear torsional spring. The model omits the catheter's distributed bending, friction, tissue contact, magnetic and mechanical hysteresis, dynamic effects, field gradients along the catheter, and detailed

magnet geometry. The 80 mT comparison field and the inferred stiffness are illustrative. Official stiffness, full baseline conditions, and the professor's tolerance and tests still need confirmation.

9 How I Would Explain Milestone 2 to My Professor

I represented the catheter tip magnet as a dipole, so its magnetic torque is $\mathbf{m} \times \mathbf{B}$. I also checked that result with a small discretized current loop using the force and torque integrals. The net force on the closed loop in a uniform field was essentially zero, and the loop torque agreed with the dipole equation to numerical precision. To estimate the tip angle, I balanced magnetic torque against a linear shaft-restoring torque and solved the resulting equation. Since the workspace did not include the shaft stiffness, I inferred an explicitly illustrative value from the approximate 30-degree baseline at an assumed 80 mT. So the model shows how field strength and stiffness affect steering, but the default baseline match is calibration, not independent proof; I need the official stiffness and baseline conditions to verify a physical result.